Overview on Cyber Security

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Cybersecurity is all about protecting your computer, phone, or any digital device from hackers and online threats. It keeps your personal information, bank details, files, and online activity safe from being stolen, damaged, or misused. By acquiring knowledge of cyber attacks and cyber security we can secure and defend ourselves from various cyber attacks like **phishing** and **DDoS** attacks.

https://www.geeksforgeeks.org/ethical-hacking/what-is-cyber-security/

Cyber security is the study of …

- Weaknesses in systems and attacks against them.
- Defending against such attacks
- Pro-actively protecting data against various attacker models

Q1: Can you define a “secure system”?

Q2: Real examples of “secure systems”?

Cyber Security

- The protection of available information or information resources from unauthorized access, attack, theft, or data damage.

- organizations must secure confidential data.
- Data in all forms must be protected.
- It minimizes business risks

Goals

Goals	Particulars
Prevention	• Preventing unauthorized access to information.
Detection	• Intrusion detection in network and unauthorized access.
Recovery	• Resilience and recover from disaster and intrusion damage for business continuity.

Risk

A concept that indicates exposure to the chance of damage or loss, and signifies the likelihood of a hazard or dangerous threat.

- Risk is associated with system, power, network, and physical losses and it can affect people, practices, and processes.

Vulnerabilities

Any condition that leaves a device open to attack.

- Improperly configured or installed hardware or software.
- Delays in applying and testing software and firmware patches.
- Untested software or firmware patches.
- Bugs in software or OSs.
- Misusing software or communication protocols.
- Poorly designed networks.
- Poor physical security.
- Insecure passwords.
- Design flaws in software or OSs.
- Unchecked user input.

Threats

Any event or action that could potentially cause damage to an asset.

- Cyber Security Threats are
- Change of Information
- Interruption of Services
- Denial of Access
- Damage to Hardware
- Damage to facilities

Attacks

A technique used to exploit a vulnerability in an application or physical computer system without the authorization to do so.

- Attacks can be in the form of
- Physical security attacks
- Software based attacks
- Social Engineering Attacks
- Attack to Web or mobile application
- Network based attack

Controls

Countermeasures that you need to put in place to avoid, mitigate, or counteract security risks due to threats and attacks.

- Preventive Controls – Biometric Access Controls
- Detection Controls – Camera
- Correction Controls – Alert System

Types of Controls

Prevention controls: prevent a threat or attack from exposing a vulnerability.

Detection controls: discovers if a threat or vulnerability in network.

Correction controls: Mitigate the consequences of a threat or attack.

Types of Threats

- Attack using social engineering
- Impersonation (an act of pretending to be another person for the purpose of entertainment or fraud.)
- Phishing (a scam where attackers deceive people into revealing sensitive information or installing malware such as viruses, worms, adware, or ransomware.)
- Using Exploits and malicious code
- Virus, Adware, Spyware, Trojan horses,
- Ransomware (a type of malicious software designed to block access to a computer system until a sum of money is paid.)
- Man in the middle attack
- DDoS attack – Distributed denial of service attack

Types of Threats.....

- IP and MAC Spoofing
- ARP Poisoning or Spoofing

A network-based attack where an attacker with access to the target network redirects an IP address to the MAC address of a computer that is not the intended recipient.

- DNS Poisoning or Spoofing

A network-based attack where an attacker exploits the open nature of DNS to redirect a domain name to a different IP address. Also known as **DNS spoofing**.

Software Attacks

Any attack that targets software resources, including operating systems, applications, services, protocols, and files carried out using software or scripts comes in this category.

Application Attacks

Application attack: A software attack that targets web-based and other client-server applications.

Attack	Description
Cross-site scripting (XSS)	• Injects malicious scripts into trusted websites. • Scripts to run when a user visits the site.
injection attacks	• SQL injection.
Buffer overflow	• Exploits fixed data buffer sizes in a target piece of software.